Answer

L 10 : PROBABILITIES (PART I)

In the case of an activity where all the possible outcomes are equally likely, the probability of an event E, denoted by $P(E)$, is given by

$$P(E) = \frac{\text{number of outcomes favourable to the event } E}{\text{total number of possible outcomes}}$$

For example, consider the event of throwing a dice and obtaining a number less than 3. Among the 6 equally likely possible outcomes, only 2 outcomes: '1' and '2', are favourable to the event.

$\therefore \quad P(\text{obtaining a number less than 3}) = \frac{2}{6} = \frac{1}{3}$

Therefore, the probability of getting a number less than 3 is $\frac{1}{3}$.

1. 'PNEUMONOULTRAMICROSCOPICSILICOVOLCANOCONIOSIS' is a 45-letter word known as the longest word published in a major English dictionary. A letter is chosen at random from this word. Given that there are 9 'O's in the word, find the probability of getting an 'O'.

$$P(\text{getting an 'O'}) = \frac{9}{45} = \underline{\underline{\frac{1}{5}}}$$

2. A letter is chosen at random from the word 'WONDERLAND'. Find the probabilities of getting
   (a) a 'D',
   (b) a vowel (響音).

(a) Among the 10 letters, there are 2 letters 'D'

$\therefore \quad P(\text{getting a 'D'}) = \frac{2}{10} = \underline{\underline{\frac{1}{5}}}$

(b) Among the 10 letters, there are 3 vowels.

$\therefore \quad P(\text{getting a vowel}) = \underline{\underline{\frac{3}{10}}}$

3. If a letter is chosen at random from the word 'MATHEMATICS', find the probabilities of getting
   (a) a 'T',
   (b) an 'R'.

(a) Among the 11 letters, there are 2 letters 'T'.

$\therefore \quad P(\text{getting a 'T'}) = \underline{\underline{\frac{2}{11}}}$

(b) Among the 11 letters, none of the letters is 'R'.

$\therefore \quad P(\text{getting an 'R'}) = \frac{0}{11} = \underline{\underline{0}}$

4. A dice is thrown. Find the probabilities of getting
   (a) the number '2' or '5',
   (b) an even number.

(a) Number of favourable outcomes = 2

$\therefore \quad P(\text{'2' or '5'}) = \frac{2}{6} = \underline{\underline{\frac{1}{3}}}$

(b) Number of favourable outcomes = 3

$\therefore \quad P(\text{an even number}) = \frac{3}{6} = \underline{\underline{\frac{1}{2}}}$

5. If a dice is thrown, find the probabilities of getting a number which is
   (a) greater than 3,
   (b) equal to 8,
   (c) less than 7.

(a) Number of favourable outcomes = 3

$\therefore \quad P(\text{greater than } 3) = \frac{3}{6} = \frac{1}{2}$

(b) Number of favourable outcomes = 0

$\therefore \quad P(\text{equal to } 8) = \frac{0}{6} = 0$

(c) Number of favourable outcomes = 6

$\therefore \quad P(\text{less than } 7) = \frac{6}{6} = 1$

6. A dice is thrown. Find the probabilities of getting
   (a) an even number smaller than 4,
   (b) an even number or a number smaller than 4.

(a) There is only 1 favourable outcome, i.e. 2.

$\therefore \quad P\,(\text{an even number smaller than } 4) = \frac{1}{6}$

(b) There are 5 favourable outcomes, i.e. 1, 2, 3, 4, 6.

$\therefore \quad P(\text{an even number or a number smaller than } 4) = \frac{5}{6}$

7. An integer is chosen at random between 1 and 30 inclusive. Find the probabilities that the integer is
   (a) a multiple of 6,
   (b) between 2 and 22 inclusive.

(a) Number of multiples of 6 between 1 and 30 inclusive = 5

$\therefore \quad P(\text{a multiple of } 6) = \frac{5}{30} = \frac{1}{6}$

(b) There are 22 – 2 + 1 = 21 favourable outcomes.

$\therefore \quad P(\text{between 2 and } 22) = \frac{21}{30} = \frac{7}{10}$

8. David selects a number at random from all positive 2-digit integers. Find the probabilities of getting
   (a) a number divisible by 3,
   (b) a number divisible by 5,
   (c) a number divisible by both 3 and 5.

(a) Number of favourable outcomes $= \frac{99}{3} - \frac{9}{3} = 33 - 3 = 30$

$\therefore \quad P(\text{divisible by } 3) = \frac{30}{90} = \frac{1}{3}$

(b) Number of favourable outcomes $= \frac{95}{5} - \frac{5}{5} = 19 - 1 = 18$

$\therefore \quad P(\text{divisible by } 5) = \frac{18}{90} = \frac{1}{5}$

(c) If a number is divisible by both 3 and 5, it is divisible by 15 (L.C.M. of 3 and 5).

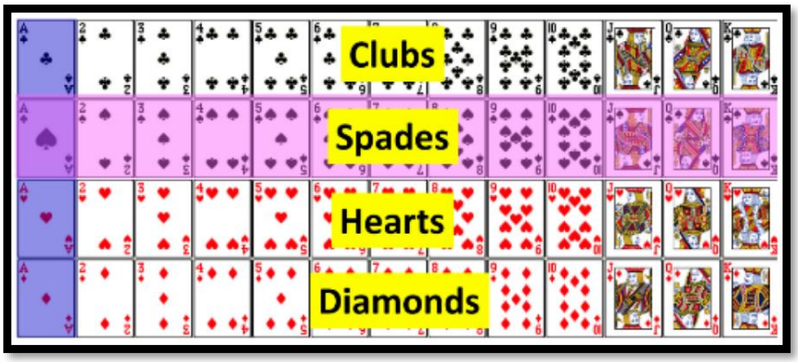
Number of favourable outcomes $= \frac{90}{15} = 6$

$\therefore \quad P(\text{divisible by both 3 and } 5) = \frac{6}{90} = \frac{1}{15}$

9. A card is drawn at random from a pack of 52 playing cards. Find the probabilities of getting

(a) a 'J',

(b) the Ace of diamonds.



(a) There are 4 'J's in a pack of 52 playing cards.

$\therefore$ Number of favourable outcomes = 4

$$\therefore\ P(\text{a 'J'}) = \frac{4}{52} = \underline{\underline{\frac{1}{13}}}$$

(b) There is only one 'A' of diamonds in a pack of 52 playing cards.

$\therefore$ Number of favourable outcomes = 1

$$\therefore\ P(\text{Ace of diamonds}) = \underline{\underline{\frac{1}{52}}}$$

10. A card is drawn at random from a deck of 52 playing cards. Find the probabilities of drawing

(a) an Ace or an '8',

(b) a black face card,

(c) a diamond or a 'K'.

(a) There are 4 Aces and 4 '8's in a deck of 52 playing cards.

$\therefore$ Number of favourable outcomes = 4 + 4 = 8

$$\therefore\ P(\text{an Ace or an '8'}) = \frac{8}{52} = \underline{\underline{\frac{2}{13}}}$$

(b) There are 12 face cards in a deck of 52 playing cards and half of them are black cards.

Altogether there are 6 black face cards in a deck of 52 playing cards.

$\therefore$ Number of favourable outcomes = 6

$$\therefore\ P(\text{black face card}) = \frac{6}{52} = \underline{\underline{\frac{3}{26}}}$$

(c) There are 13 diamonds and 4 'K's in a deck of 52 playing cards, but the diamond 'K' is both a diamond and a 'K'.

$\therefore$ Number of favourable outcomes $= 13 + 4 - 1 = 16$

$$\therefore\ P(\text{a diamond or a 'K'}) = \frac{16}{52} = \underline{\underline{\frac{4}{13}}}$$

11. A card is drawn randomly from a deck of 52 playing cards. Find the probabilities that the card drawn is

(a) an Ace or a face card,

(b) not a spade or a 'K'.

(a) There are 4 Aces and 12 face cards in a deck of 52 playing cards

$\therefore$ Number of favourable outcomes $= 4 + 12 = 16$

$$\therefore\ P(\text{an Ace or a face card}) = \frac{16}{52} = \underline{\underline{\frac{4}{13}}}$$

(b) There are 13 spades and 4 'K's in a deck of 52 playing cards, while the spade 'K' is both a spade and a 'K'.

Number of cards that are not a spade or a 'K' $= 52 - (13 + 4 - 1) = 36$

$$\therefore\ P(\text{not a spade or a 'K'}) = \frac{36}{52} = \underline{\underline{\frac{9}{13}}}$$

12. A bag contains k green beans and 8 red beans. If a bean is randomly drawn from the bag, the probability of drawing a red bean is $\frac{2}{7}$. Find the value of k.

Total number of beans in the bag $= k + 8$

From the question, we have

$$\frac{8}{k+8} = \frac{2}{7}$$

$$2k + 16 = 56$$

$$2k = 40$$

$$k = \underline{\underline{20}}$$

Unit Education

13. There are 33 male staff and n female staff in a company. If a staff is randomly selected from the company, the probability of selecting a female staff is $\frac{6}{17}$. Find the value of n.

Total number of staff in the company $= n + 33$

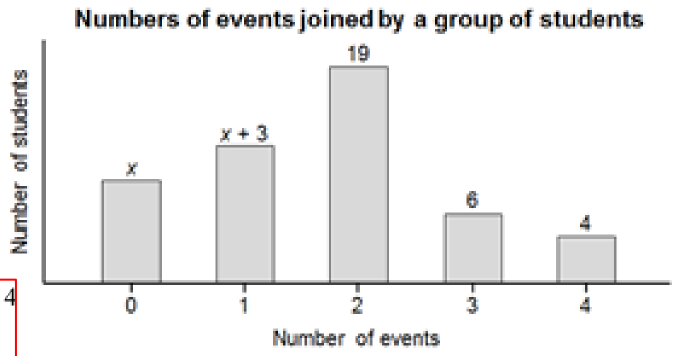
From the question, we have

$$\frac{n}{n+33} = \frac{6}{17}$$
$$17n = 6n + 198$$
$$11n = 198$$
$$n = \underline{\underline{18}}$$

14. The bar chart below shows the distribution of the numbers of events joined by a group of students on the Sport Day. If a student is randomly selected from the group, the probability that the student joins one event is $\frac{6}{25}$.

(a) Find the value of x.

(b) If a student is randomly selected from the group, find the probability that the student joins more than 2 events.



(a) Total number of students in the group $= x + (x+3) + 19 + 6 + 4$

$= 2x + 32$

$\because$ The probability that the student joins one event is $\frac{6}{25}$.

$$\therefore \quad \frac{x+3}{2x+32} = \frac{6}{25}$$
$$25x + 75 = 12x + 192$$
$$13x = 117$$
$$x = \underline{\underline{9}}$$

(b) Total number of students in the group $= 2 \times 9 + 32 = 50$

Number of students who joins more than 2 events $= 6 + 4 = 10$

$$\therefore \quad P(\text{joins more than 2 events}) = \frac{10}{50} = \underline{\underline{\frac{1}{5}}}$$

15. Three types of drinks, lemon tea, apple juice and orange juice, are packed in cartons and put on a table. The number of cartons of lemon tea is 20 more than that of apple juice while the number of cartons of orange juice is half of that of lemon tea. If a carton of drink is selected at random, the probability of getting a carton of orange juice is $\frac{2}{9}$. How many cartons of lemon tea are there on the table?

Let n be the number of cartons of lemon tea.

Then the number of cartons of apple juice is $(n-20)$

and the number of cartons of orange juice is $\frac{1}{2}n$.

Total number of cartons of drinks $= n + (n-20) + \frac{1}{2}n = \frac{5}{2}n - 20$

$\because$ Probability of getting a carton of orange juice $= \frac{2}{9}$

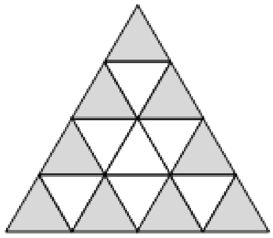
$$\therefore \quad \frac{\frac{1}{2}n}{\frac{5}{2}n - 20} = \frac{2}{9}$$
$$\frac{n}{5n-40} = \frac{2}{9}$$
$$9n = 10n - 80$$
$$n = \underline{\underline{80}}$$

Geometric Probability

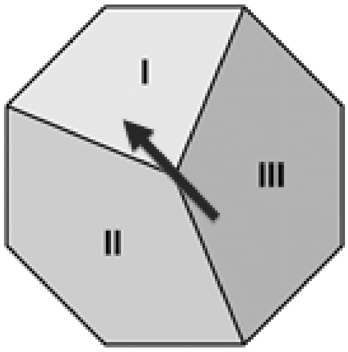
16. The figure shows a dartboard which is made up of 16 identical equilateral triangles. Philip throws a dart at random and the dart hits the dartboard. Find the probability that the dart hits the shaded region.

Among the 16 identical triangles, 9 are shaded.

$\therefore \quad P(\text{hits the shaded region}) = \underline{\underline{\dfrac{9}{16}}}$



17. The figure shows a lucky wheel in the shape of a regular octagon. Chris spins the lucky wheel once. Find the probabilities that the pointer
(a) stops in region I,
(b) does not stop in region II.



(a) The lucky board is in the shape of a regular octagon.
Divide the octagon into 8 equal parts. Among the 8 parts, 2 are in region I.

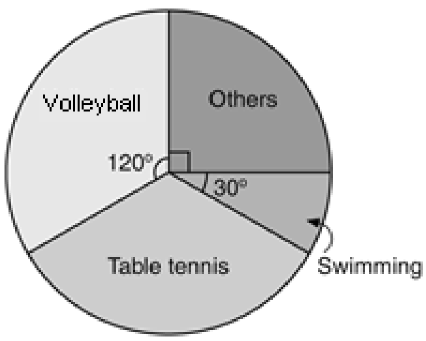
$\therefore \quad P(\text{stops in region I}) = \dfrac{2}{8}$

$= \underline{\underline{\dfrac{1}{4}}}$

(b) Among the 8 parts, 5 are not in region II.

$\therefore \quad P(\text{does not stop in region II}) = \underline{\underline{\dfrac{5}{8}}}$

18. The pie chart shows the distribution of the favourite sports of a group of 180 students. If a student is chosen from the group at random, find the probabilities that his/her favourite sport is
(a) swimming or volleyball,
(b) table tennis.



(a) Sum of angles at the centre of the sectors representing 'swimming' and 'volleyball'
$= 30° + 120° = 150°$

$\therefore \quad P(\text{swimming or volleyball}) = \dfrac{150°}{360°}$

$= \underline{\underline{\dfrac{5}{12}}}$

(b) Angle at the centre of the sector representing 'table tennis'
$= 360° - 120° - 90° - 30°$
$= 120°$

$\therefore \quad P(\text{table tennis}) = \dfrac{120°}{360°}$

$= \underline{\underline{\dfrac{1}{3}}}$

19. The figure shows a circular dartboard. The areas of the 3 sectors B, C and D are in the ratio 2 : 1 : 1. If a dart is thrown at random and hits the dartboard, find the probability that the dart hits the sector C.

Let r be the radius of the dartboard.

$\because$ The size of a round angle is 360°.

$\therefore$ Angle of sector B + angle of sector C + angle of sector $D = 360° - 90°$

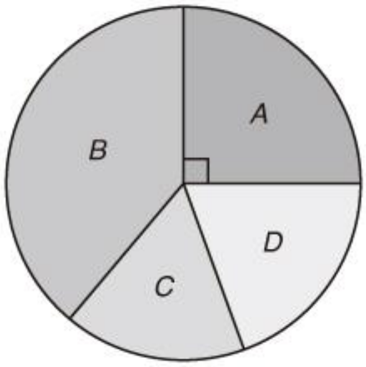
$= 270°$

Angle of sector $C = 270° \times \dfrac{1}{2+1+1}$

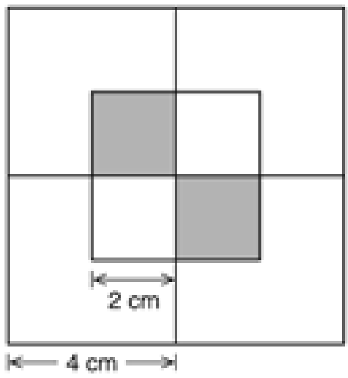
$= 67.5°$

$\therefore$ $P(\text{sector } C) = \dfrac{67.5°}{360°}$

$= \dfrac{3}{16}$



20. As shown in the figure, a dartboard is formed by 4 squares each of side 4 cm. Another 4 squares each of side 2 cm are drawn at the centre of the dartboard. If a dart is thrown at random and hits the dartboard, find the probability that it hits one of the shaded regions.



Area of the dartboard $= (4+4)^2 \text{ cm}^2$

$= 64 \text{ cm}^2$

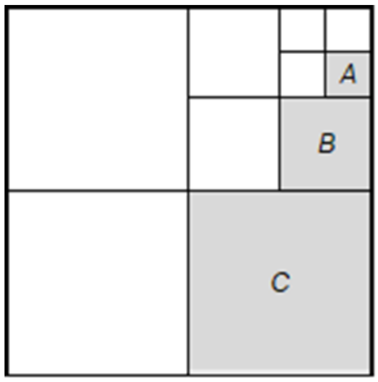
Total area of the shaded regions $= 2^2 \times 2 \text{ cm}^2$

$= 8 \text{ cm}^2$

$\therefore$ $P(\text{hits one of the shaded regions}) = \dfrac{8 \text{ cm}^2}{64 \text{ cm}^2}$

$= \dfrac{1}{8}$

21. As shown in the figure, a dartboard is formed by 4 squares each of side 4 cm. Another 4 squares each of side 2 cm are drawn at the centre of the dartboard. If a dart is thrown at random and hits the dartboard, find the probability that it hits one of the shaded regions.



Area of the dartboard $= 4 \times 4 \times 4 \times$ area of square A

$= 64 \times$ area of square A

Area of square $B = 4 \times$ area of square A

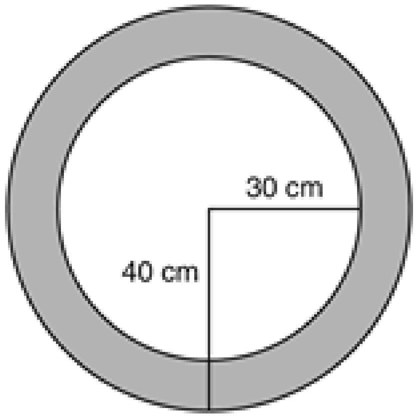
Area of square $C = 4 \times$ area of square B

$= 16 \times$ area of square A

$\therefore$ $P(\text{shaded region}) = \dfrac{(16+4+1) \times \text{area of square } A}{64 \times \text{area of square } A}$

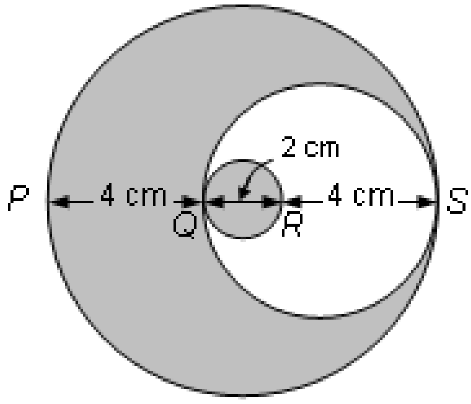
$= \dfrac{21}{64}$

22. The dartboard shown in the figure is made up of two concentric circles of radii 30 cm and 40 cm respectively. Kenny throws a dart at random and the dart hits the dartboard. Find the probability that the dart hits the shaded region.



$$\text{Area of the dartboard} = \pi \times 40^2 \text{ cm}^2$$
$$= 1600\pi \text{ cm}^2$$
$$\text{Area of the shaded region} = (\pi \times 40^2 - \pi \times 30^2) \text{ cm}^2$$
$$= (1600\pi - 900\pi) \text{ cm}^2$$
$$= 700\pi \text{ cm}^2$$
$$\therefore \quad P(\text{hits the shaded region}) = \frac{700\pi \text{ cm}^2}{1600\pi \text{ cm}^2}$$
$$= \underline{\underline{\frac{7}{16}}}$$

23. The figure shows a dartboard with a pattern formed by 3 different sizes of circles. PS, QR and QS are the diameters of the circles. Mable throws a dart at random and the dart hits the dartboard. Find the probability that the dart hits the shaded region.

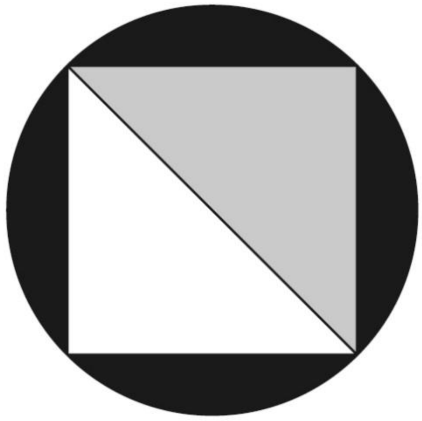


Let circles A, B and C denote the largest circle, the second largest circle and the smallest circle respectively.

$$\text{Area of circle } A = \pi \times \left(\frac{4+2+4}{2}\right)^2 \text{ cm}^2$$
$$= 25\pi \text{ cm}^2$$
$$\text{Area of circle } B = \pi \times \left(\frac{4+2}{2}\right)^2 \text{ cm}^2$$
$$= 9\pi \text{ cm}^2$$
$$\text{Area of circle } C = \pi \times \left(\frac{2}{2}\right)^2 \text{ cm}^2$$
$$= \pi \text{ cm}^2$$
$$\therefore \quad P(\text{hits the shaded region}) = \frac{(25\pi - 9\pi + \pi) \text{ cm}^2}{25\pi \text{ cm}^2}$$
$$= \frac{17\pi}{25\pi}$$
$$= \underline{\underline{\frac{17}{25}}}$$

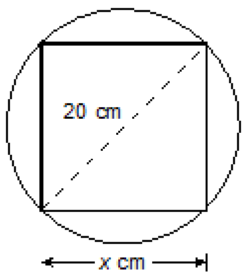
24. The figure shows a circular dartboard of diameter 40 cm. Two congruent triangles are drawn to form a square inscribed in the dartboard. If a dart is thrown at random and hits the dartboard, find the probability that it hits the grey region. (Express your answer in terms of π.)

$$\text{Area of the dartboard} = \pi \times \left(\frac{40}{2}\right)^2 \text{ cm}^2$$
$$= 400\pi \text{ cm}^2$$
$$\text{Area of the grey region} = \frac{1}{2} \times 40 \times \left(\frac{1}{2} \times 40\right) \text{cm}^2$$
$$= 400 \text{ cm}^2$$
$$\therefore \quad P(\text{grey region}) = \frac{400 \text{ cm}^2}{400\,\pi \text{ cm}^2}$$
$$= \underline{\underline{\frac{1}{\pi}}}$$



25. A circular dartboard with the diameter of 20 cm is shown with a square inscribed in it. Suppose a dart hits a point randomly on the circular dartboard without hitting any boundaries, find the probability that it hits the square region. (Express your answer in terms of π.)

Let x cm be the length of each side of the square.



According to Pythagoras' theorem,

$$x^2 + x^2 = 20^2$$
$$2x^2 = 400$$
$$x^2 = 200$$

$$\text{Area of the square} = x^2 \text{ cm}^2$$
$$= 200 \text{ cm}^2$$
$$\text{Area of the dartboard} = \pi\left(\frac{20}{2}\right)^2 \text{ cm}^2$$
$$= 100\pi \text{ cm}^2$$
$$\therefore \quad P(\text{hitting the square region}) = \frac{200}{100\pi}$$
$$= \underline{\underline{\frac{2}{\pi}}}$$